University of Central Punjab

Faculty of Information Technology & Computer Sciences

Data Structures, Spring 2026

Assignment 1

Total Marks: 50	**Deadline: 20 - 03 - 2026**	**Section:**
Name:	**Reg. #:**	

CLO	CLO Statement	Taxonomy Level	PLO
1	Understand the basic knowledge of data structures and algorithms.	C2	**2**

Instructions: (5 marks will be deducted for not following the instructions)

- You have to submit Assignment 1 via Portal. Do individually.
- Quiz will be taken on the basis of assignment.
- In case of plagiarism, ***straight ZERO*** will be awarded.
- Attempt all questions in sequence. **Attach this title page as a front page of assignment**.
- Assignments will not be accepted on email and late submissions of assignments are strictly prohibited.
- Attach output screen shot of each question.

Answers Clarity: Justify your answer where needed.

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Question # 1: Perform a binary search on the following array searching for the number 18.
10 Marks

arr = {12, 77, 87, 89, 100, 117, 125, 189, 235, 529, 1000}.

Tasks:

1. Count the total number of comparisons.
2. Complete the following table:

Iteration	Low	High	Mid	A[Mid]	Decision

Question # 2: Write a code to construct an array using the **ASCII values of the letters in your first name**. **10 Marks**

Example:

Name: **ALI**

A = 65
L = 76
I = 73

Tasks:

1. Insert the **ASCII value of your last name's first letter** at position 2.
2. Delete the element at position 3.
3. Replace the last element with **100**.

For each step show:

- Array before operation
- Array after operation
- Number of elements shifted

Question # 3: Write a code that use the first 5 digits of your student ID **as stack elements.**

Implement a **stack using an array**. The stack follows the **Last-In-First-Out (LIFO)** principle and has a **maximum capacity of 5 elements**. **10 Marks**

The stack is represented using global variables:

```
#define S_capacity 5
int Stack[S_capacity];
int S_size = 0;
```

Write the following functions:

1. **stack_print()**
   - Print all elements in the stack.
   - If the stack is empty, print a message indicating it.
2. **stack_push(int value)**
   - Insert a value into the stack.
   - Return **0 if successful**.
   - Return **5 if the stack is full**.
3. **stack_pop()**
   - Remove and return the top element from the stack.s
   - Return **-1 if the stack is empty**.

Sample Output:

```
Stack content: Stack is empty.

Pushing 1...   SUCCESS    Stack content: 1
Pushing 2...   SUCCESS    Stack content: 1 2
Pushing 3...   SUCCESS    Stack content: 1 2 3
Pushing 4...   SUCCESS    Stack content: 1 2 3 4
Pushing 5...  SUCCESS     Stack content: 1 2 3 4 5
Pushing 6...   FAIL       Stack content: 1 2 3 4 5

Popping from stack...  DATA: 5   Stack content: 1 2 3 4
Popping from stack...  DATA: 4   Stack content: 1 2 3
Popping from stack...  DATA: 3   Stack content: 1 2
Popping from stack...  DATA: 2   Stack content: 1
Popping from stack...  DATA: 1   Stack content: Stack is empty.
Popping from stack...  FAIL      Stack content: Stack is empty.
Popping from stack...  FAIL      Stack content: Stack is empty.
```

Question # 4: Convert the following **infix expression** into **prefix and postfix notation**.

10 Marks

A + B * (C ^ D - E) ^ (F + G * H) – I

Tasks:

1. Show the **operator stack at each step**.
2. Show the **output expression formation**.
3. Write the **final prefix and postfix expressions**.

Question # 5: A university database system performs these operations: **10 Marks**

Task	Description
T1	Retrieve student record using index
T2	Search a student in an unsorted list
T3	Compare all students with each other to detect duplicate profiles
T4	Search a sorted list using Binary Search

Tasks:

1. Assign the most appropriate **time complexity** to each task from the following: **O(1), O(N), O(N^2), O($\log_2 N$)**. Justify your answer briefly.
2. Arrange the algorithms **from most efficient to least efficient** when the number of records becomes extremely large.
3. The university plans to increase its database to **several million records**. Recommend **two tasks that should be optimized or redesigned** to improve performance and explain why.
4. Suggest **one alternative approach or data structure** that could improve the performance of the least efficient task.